



ELSEVIER

Journal of Non-Crystalline Solids 311 (2002) 107–117

JOURNAL OF
NON-CRYSTALLINE SOLIDS

www.elsevier.com/locate/jnoncrysol

Molecular dynamics simulation of $\text{La}_2\text{O}_3\text{--Na}_2\text{O--SiO}_2$ glasses. II. The clustering of La^{3+} cations

Byeongwon Park, L. René Corrales *

*Environmental Molecular Sciences Laboratory, Pacific Northwest National Laboratory, P.O. Box 999, 902 Batelle Blvd., K8-91,
Richland, WA 99352, USA*

Received 20 July 2001; received in revised form 28 February 2002

Abstract

Clustering of high-field strength rare-earth ions in silicate glasses has been experimentally observed for a wide range of concentrations. Clustering has also been observed by molecular dynamics (MD) computer simulations over a range 1–10 mol% in soda silicate glasses. Although there have been numerous experimental studies, atomic-level details of the mechanisms that lead to clustering remain unclear. Coupling experiment with MD simulations is essential to uncovering the factors that lead to clustering. In this work, MD computer simulations are used to verify that clustering found in previous MD simulations is not an artifact of the simulation method. This work also provides clues as to the mechanism of atomic-level clustering.

© 2002 Elsevier Science B.V. All rights reserved.

1. Introduction

The certainty of the role of La ions as a network modifier in soda silicate glasses has been well established by both experiment [1] and simulations [2]. A network modifier is a metal oxide that inserts into a silica network by breaking a Si–O bond to form a pair of oxygen anions where each oxygen atom is bonded to only one silicon atom. Thus, the network is effectively depolymerized. Molecular dynamics (MD) simulations have revealed that La

ions cluster readily [2]. Clustering is defined by the existence of La–O–La linkages consistent with experimental interpretation of related systems [1,3–5]. Evidence for La clustering was found to occur in the systems simulated over a range of 1–10 mol% of La_2O_3 [2], in agreement with experimental findings of related systems [6]. The simulation results show that clustering of the La cations leads to a structure in which aggregates are inhomogeneously dispersed throughout the glass. This is different than that found for the alkali ions in silicate glasses where channeling or fingering of the ions occurs [7,8]. The difference is likely due to the high valence of the La cation that leads to a tightly bound first nearest-neighbor shell. The extent to which the clustering of high-field strength rare-earth ions is real, and not an artifact of the

* Corresponding author. Tel.: +1-509 376 6608; fax: +1-509 376 0420.

E-mail address: rene.corrales@pnl.gov (L.R. Corrales).

MD simulation procedure or experimental interpretation requires a proper description of the microstructure. This can be accomplished by corroboration between experiment and simulation [9]. In previous MD simulations, henceforth referred to as paper I [2], confidence in the potential model was established by analyzing structural results that were directly compared to NMR studies [1] for the same glass mixtures. In that work, EXAFS spectra and neutron diffraction patterns were also predicted, for which no experimental data currently exists, that can lead to further establishing the reliability of the potential model. In this work, the propensity of aggregation is characterized by designing specific MD simulation experiments that also explore the influence of the network structure on the La ion distribution. This is done by providing different initial configurations, as described below, then carrying out both glass forming and annealing simulations. Such computer experiments can also provide insight into the driving forces that lead to clustering.

The paper is organized as follows. In the following section, a brief description of the computer simulation procedure and experiments are provided. In Section 3, the results and analysis of the simulations are given with a discussion following in Section 4. A conclusion is presented in Section 5.

2. Computer simulation methodology

The potential model and MD simulation procedure are identical to that of paper I. The MD simulation code used in this study was DL_POLY from Daresbury Laboratory, UK [10]. A rigid-ion model is used where the ions are represented by their formal charges; Na^{1+} , Si^{4+} , La^{3+} , O^{2-} . The long-range electrostatic potential was evaluated by

the Ewald summation method [11] to a precision of 1×10^{-6} and a cutoff distance set at 12.0 Å. The short-range cutoff distance was set at 7.6 Å. The potential parameters for SiO_2 are based on Vesal's model that includes three-body potentials for Si–O–Si and O–Si–O interactions [12]. The potential for the La–O and Na–O interactions also use the modified Buckingham potential where the La–O parameters were initially fit to crystalline La_2O_3 . The parameters for La–O and Na–O are given in paper I. The length of each time step was 2×10^{-15} s and the integration was done using the Verlet Leapfrog algorithm [13].

The composition of the simulated glass, the total number of atoms, and the simulation box size used in this work is given in Table 1, which is equivalent to the Na3S4 mixture studied of paper I and corresponds to a concentration of 4 mol%. All the systems simulated in the original study showed a strong propensity to cluster, beyond that which would be statistically expected for this concentration. This system was selected because its concentration is intermediate of the range previously simulated. The structural properties, such as the PDF, coordination number distribution, and bond angle distribution for each of the simulations were obtained over the final room temperature configurations, and are presented in detail below.

Four distinct simulation experiments were carried out to determine the persistence and propensity of clustering in this system. These simulations are designed to verify that clustering is not an artifact of the simulation and to clarify the role of the melt-quench process on the clustering of La in the silicate glass structure. Simulation 1 is our reference simulation for the composition of Na3S4 of Table 1. As reported in paper I, the La, and corresponding O atoms, were distributed randomly in the initial configuration of a previously simulated

Table 1

Composition, number of atoms, simulation box size and the theoretical ratio of number of non-bridging oxygen (NBO) to number of silica tetrahedron (T) of the simulated glass

System	Concentration			Number of atoms				Atoms in box	Box length (Å)	NBO/T
	La_2O_3	SiO_2	Na_2O	La	Na	Si	O			
Na3S4	4.0	72.0	24.0	40	240	360	900	1540	27.313	1.00

soda silicate glass. Here random means that the distribution of La was not controlled. In principal, clusters can occur in a statistical distribution. The tails (low probability region) of a statistical distribution are expected to be made up of a homogeneous distribution or of a single cluster. The system was melted at 12,000 K to remove any possible memory effects and then taken through a step-cooling glass-forming procedure to room temperature. The details of the melt-quench process and the analysis of glass structure can be found in paper I. In the simulation of paper I, the ions (La, Na, Si and O) displaced a distance of at least half of the simulation box length during the initial melting at high temperature (12,000 K) where the heavier La ions diffused the slowest, and, hence, the reason for the high melt temperature.

In simulation 2 the initial configuration consists of uniformly distributed La ions in a cubic array with an interatomic distance of 7.8 Å. This configuration is then taken through the same step-cooling glass-forming procedure identical to that of simulation 1. Simulation 3 has the same initial configuration as simulation 2, except that a frozen constraint is applied to the uniformly distributed La cations. In this simulation, the La ions were held frozen during the step-cooling glass-forming process where all the Si, O, and Na ions were provided the thermal energy of the target temperatures and allowed to rearrange about the frozen La ion configuration.

Simulation 4 consists of two annealing experiments to find the natural tendency of La clustering. In these simulations, the final room temperature configuration of simulation 3 was annealed (at 5000 and 3000 K) with the frozen atom constraint for the La ions removed. The initial structure was allowed to relax to lower energy configurations at constant temperature. The annealing simulations are run in the canonical ensemble (NVT) using an Evans thermostat [14] with a total simulation time of 220 ps.

3. Results and analysis

For reference, the La–La PDF for each of the compositions studied in paper I [2] are shown in

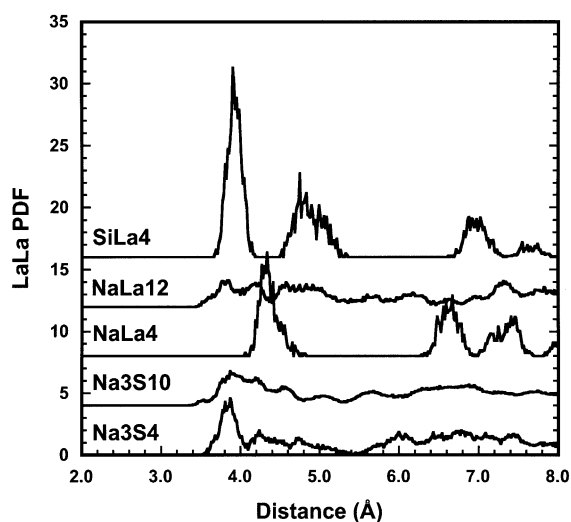


Fig. 1. The La–La PDFs of the glass mixtures simulated in paper I [2] for the corresponding compositions shown in Table 2. All the mixtures show clustering regardless of the concentration of La ions or the presence of Na ions.

Table 2
Composition of simulated glasses shown in Fig. 1 as carried out in Ref. [2]

System	Concentration (mol%)		
	La ₂ O ₃	Na ₂ O	SiO ₂
Na3S4	4.0	24.0	72.0
Na3S10	10.0	22.5	67.5
NaLa4	1.4	21.6	77.1
NaLa12	4.4	14.1	81.5
SiLa4	1.7	0.0	98.3

Fig. 1 (Table 2). The average La–La nearest-neighbor distance is given by the first peak and is around 3.9 Å for all the simulated glasses, with the exception of the NaLa4 mixture which has the first peak shifted to about 4.2 Å. At lower concentrations of La₂O₃ (<2%), the La–La PDF show relatively sharp peaks. The addition of Na shifts the first peak from 3.9 Å for SiLa4 to about 4.2 Å for NaLa4. In addition, the second peak of SiLa4 centered at about 4.8 Å disappears upon addition of Na ions. For concentrations of La₂O₃ at about 4% (NaLa12 and Na3S4), the addition of Na ions significantly reduces the sharpness of the first peak. Raising the concentration of La to 10% (Na3S10) does not significantly change the shape of the first peak, although there are changes in the shape of

the PDF after the first peak. It is intriguing that the addition of La ions to a soda silicate glass has almost no effect on the shape of the Na–Na PDF (see paper I), whereas the addition of Na ions to a lanthanum silicate glass leads to strong changes in the La–La PDF. In spite of what appear to be structural differences between these glass mixtures, La polyhedra clustering as determined by the presence of La–O–La bond linkages are found for all the glass mixtures. Hence, only one intermediate mixture was used in this study.

In Fig. 2, only the La–O polyhedra of the final configuration of the Na₃S₄ simulated glass from paper I are shown, all other atoms have been removed for clarity. The first nearest-neighbor O atoms are chosen within a cutoff distance of 3.2 Å as determined in paper I. Fig. 2 shows the clustering of La–O polyhedron, with La–O–La linkages, as well as some isolated La–O polyhedra. In simulation 1, the La ions were initially distributed *randomly* in the simulation box and melted at 12,000 K to obtain a homogeneous melt, and then the system step cooled to form glass. The La ions

in soda silicate glasses show a strong tendency to cluster. The direct La–O–La linkages were found to have a La–La distance varying between 4 and 5 Å. In the initial simulations of paper I, it was not possible to ascertain when the clustering of the La polyhedra occurred, or to what extent the initial configuration influenced the formation of such clusters. It is conceivable that the clustering is simply a result of the statistical distribution, where nearest-neighbor polyhedra are not excluded. Hence, further investigation is warranted.

To this end, the initial configuration of La atoms were controlled by uniformly distributing them in the simulation box, as shown in Fig. 3. Two simulations were then carried out, the first to determine the role of the initial configuration and the second to determine the energetics and structure that would be obtained if the La ions remained uniformly distributed. In simulation 2, the uniformly distributed La structure was taken through the melt-quench glass-forming process where all the ions were allowed to move, as in simulation 1. The resulting structure has features

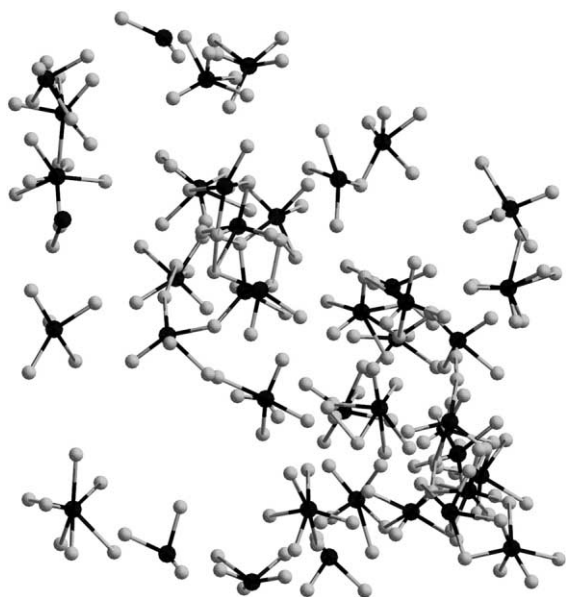


Fig. 2. The final configuration of simulation 1, as determined for the simulations of paper I [2]. The composition of this mixture is given in Table 1. Only the polyhedra consisting of the La ions and their first nearest-neighbor O ions determined by a cutoff of 3.2 Å are shown for clarity.

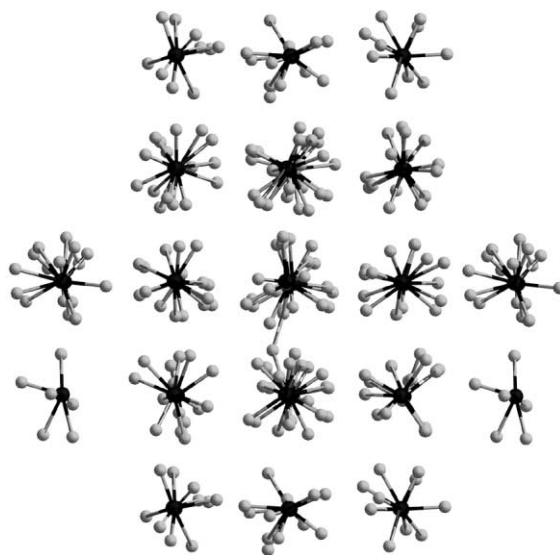


Fig. 3. The initial configuration for simulations 2, 3 and 4 is that where the La ions are uniformly distributed. The distance between them is the maximum possible separation distance for this concentration of La ions. In simulation 3, the Si, Na and O atoms were taken through the melt-quench glass-forming process while keeping the La ions frozen. This figure shows the La polyhedra resulting from simulation 3.

nearly identical to that of simulation 1. The resulting distribution of La–O polyhedra are shown in Fig. 4. It is clear that the initial configuration does lose its memory during the high-temperature melt run, as expected. Again, it was not clear from this simulation when clustering occurred. Note that the potential energy of the glass from simulation 2 is higher than that of simulation 1, even after further annealing. This is not surprising since simulation 2 represents a second possible glass structure out of an infinite set of possible structures of the glass mixture, and also the initial configuration of simulation 1 may have already had clusters formed during the initial random distribution of the ions. The size of the system is still small relative to a real system. Thus, to get a good statistical average many glass configurations of the same mixture would have to be sampled. Alternatively, a much larger glass system could be simulated. It is important to note that this method of simulating a glass structure does lead to excellent structural results, as shown in paper I, and the consistency between simulation 1 and 2 indicates

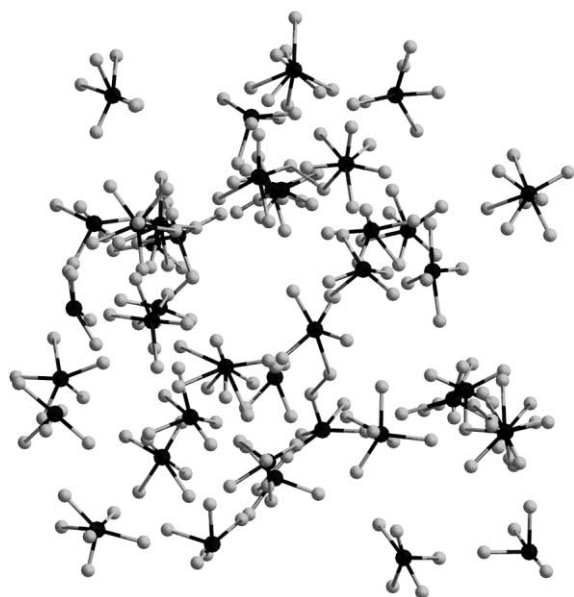


Fig. 4. The resulting distribution of the La polyhedra from simulation 2 where all the ions were allowed to diffuse during the melt-quench glass-forming process just as in simulation 1. The structural distribution of all the ions are similar to that of simulation 1.

that the sampling is within a range of the most probable structure and not at the tails (less probable states) of a typical glass structure distribution.

In simulation 3, the uniformly distributed La structure was taken through the melt-quench glass-forming process with the La atoms frozen in place and all other atoms allowed to move. The resulting structure is that of Fig. 3, which shows only the La–O polyhedra. In this simulation, the distribution of the Si, O and Na atoms did change significantly from the initial configuration. Interestingly enough, the PDFs corresponding to Si–O, Si–Si, O–O, Na–O, Na–Na and Na–Si are all very similar to those of simulations 1 and 2. Only the La–La, La–Si, La–O and La–Na curves showed differences. The similarities and differences are shown in Fig. 5 where only the cation-oxygen PDFs are shown. In addition, there is a distinct difference in the energetics of the system. The potential energy difference between simulation 1 and simulation 3 is about 38 eV at room temperature, as shown in Fig. 6. Hence, a uniform distribution is not energetically favored. Note that this is not an improbable configuration, but it is certainly a less likely configuration.

In simulations 4a and 4b, the resulting room temperature structure of simulation 3 is annealed for 220 ps at 3000 and 5000 K. The annealing at 3000 K (simulation 4a) does not lead to significant diffusion of the La ions, and hence no clustering is observed, as shown in Fig. 7. However, the annealing at 5000 K (simulation 4b) does show diffusion of the La ions and results in the appearance of clusters, as shown in Fig. 8. Although, the annealing simulation shows that the diffusion barrier temperature is somewhere between 3000 and 5000 K, it must be kept in mind that the simulation time is short relative to real experiments and that the melting temperature of most any simulation without a surface exposed is high when compared to experiment. These factors are also strongly dependent on the potential model in that the fits are made to equilibrium states and not to a complete set of physical properties. Nonetheless, these annealing simulations reveal that the system has a tendency to cluster without first forming a homogeneous melt. Hence, clustering is not due solely to a statistical distribution of the ions. There is

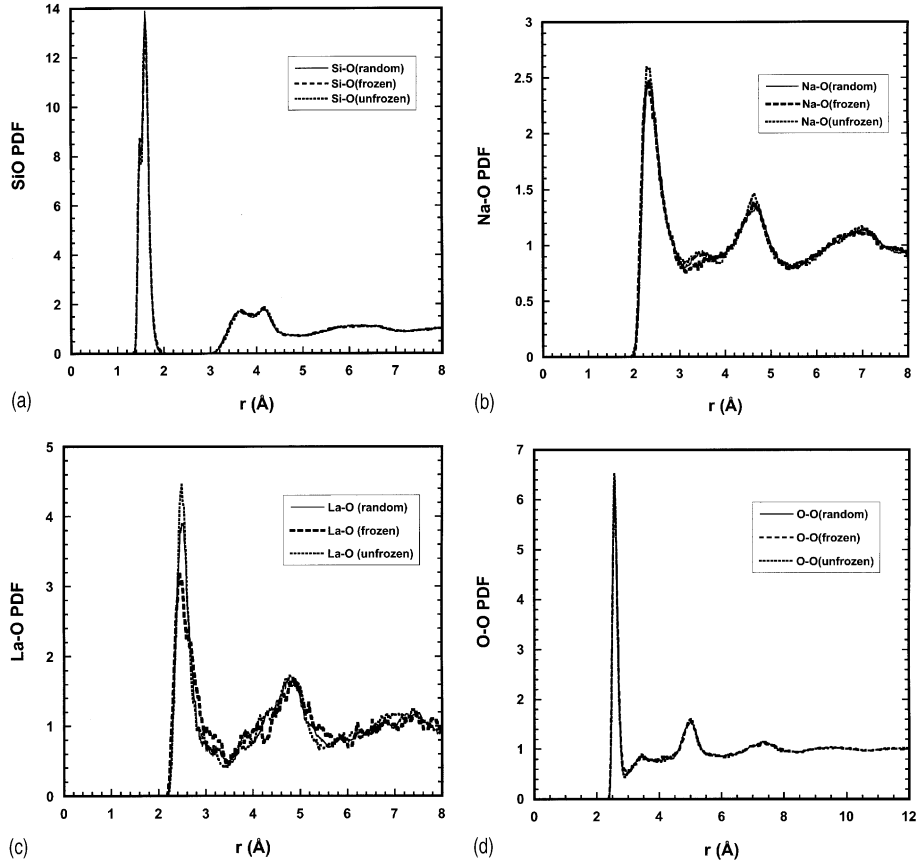


Fig. 5. The PDFs of the cations with O. Note that the Si–O and Na–O PDFs for the cases of simulation 1 (random), 2 (unfrozen) and 3 (frozen) are virtually identical. Whereas, for the La–O PDFs there is a marked difference for that of simulation 3 (frozen). The slight difference between that of simulations 1 and 2 are due to sampling a distinct glass configuration.

clearly an influence of the energies to form the La–O–La linkages and of the relaxation of network structure about the La–O polyhedron. This can be further characterized as follows.

The degree of symmetry of the cation polyhedra can be probed by calculating the dipole moments of the cation polyhedra. The smaller the dipole moment, the more symmetrically arranged are the O anions about the cation. The polyhedra consist of the cation and the nearest-neighbor O anions within a specified cutoff for each cation given by $r_{\text{La-O}} = 3.2 \text{ \AA}$, $r_{\text{Si-O}} = 2.2 \text{ \AA}$ and $r_{\text{Na-O}} = 3.0 \text{ \AA}$. The dipole moment of a cation polyhedra is defined as

$$\mu = \sum_i \vec{q}_i \cdot (\vec{r}_i - \vec{r}_{\text{CM}}) \quad (1)$$

where r_{CM} is the center of mass of the cation polyhedron. A typical distribution is shown in Fig. 9 for the Na₃S₄ glass composition. The average dipole moment for the Si polyhedra is the lowest and has a narrow distribution that implies a symmetric arrangement of the SiO₄ tetrahedron. The dipole moment of the Na polyhedra is next in magnitude with a wider distribution than that of the Si polyhedra. The dipole moment for La polyhedra has the widest distribution and has the largest magnitudes of the dipole moment because of greater variation in symmetry. Both the Na and La cations have a wider range of coordination number than that of the Si cation. The variation of the dipole moment can be used to probe changes in local structure, as shown below.

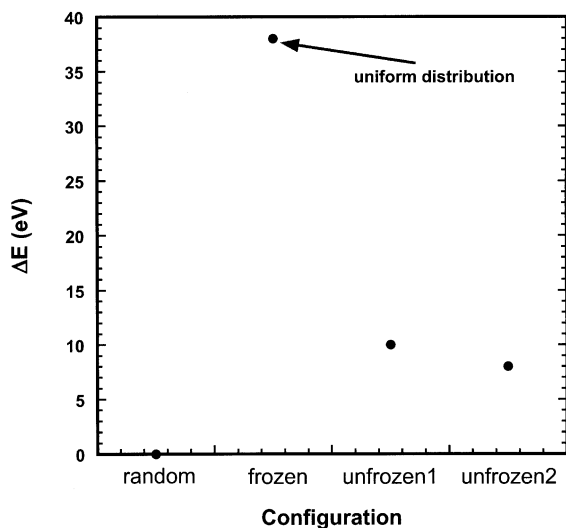


Fig. 6. The total configuration energy for simulation 1 (random), simulation 3 (frozen) and simulation 2 (unfrozen 1 and 2). The difference between simulation 1 and 3 is about 38 eV. Simulation 2 was further annealed with a lowering of the energy of about 2 eV. Differences in energy of simulation 1 and 2 are related to the initial configuration.

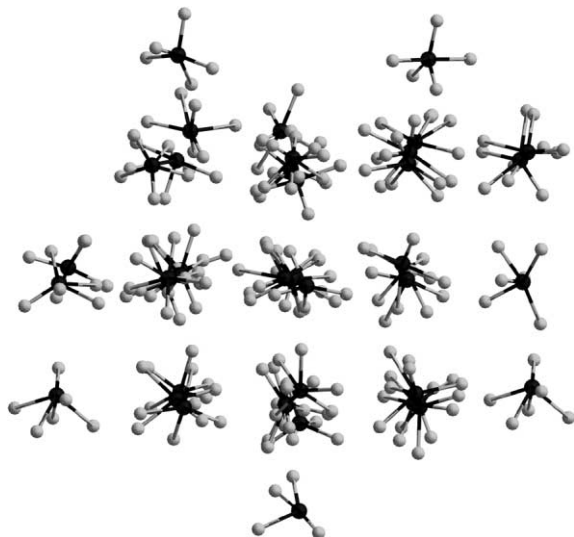


Fig. 7. Resulting configuration from simulation 4a where the configuration of simulation 3 was annealed at 3000 K and where all the ions are mobile. Small changes in the distribution are seen, but no clustering has occurred after a period of 220 ps. Longer runs may lead to clustering.

The dipole moment distribution of the La–O polyhedron for the room temperature configu-

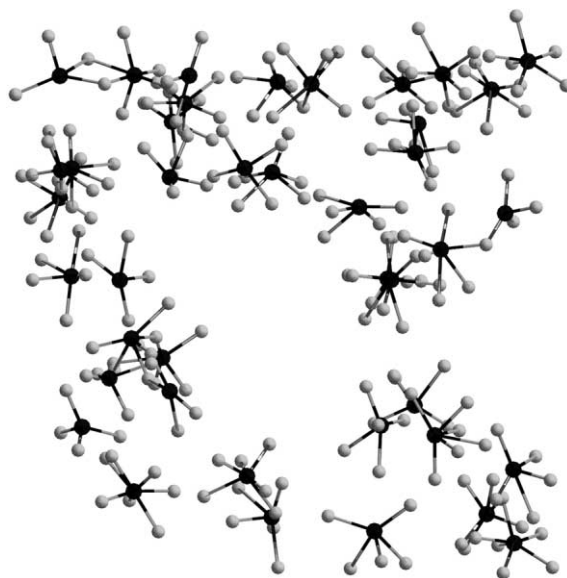


Fig. 8. Resulting configuration from simulation 4b where the configuration of simulation 3 was annealed at 5000 K and where all the ions are mobile. The La ions have shown a large degree of clustering after 220 ps.

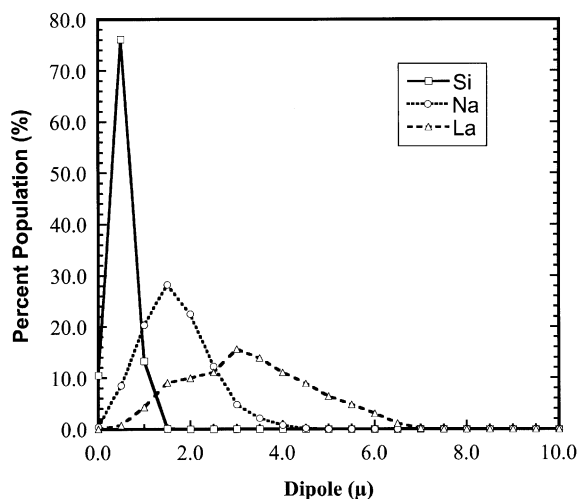


Fig. 9. The molecular dipole moment, as determined by Eq. (1), for the cation polyhedra. The first nearest-neighbor oxygen that make up the molecular polyhedron are determined by using cutoffs of $r_{\text{La-O}} = 3.2 \text{ \AA}$, $r_{\text{Si-O}} = 2.2 \text{ \AA}$ and $r_{\text{Na-O}} = 3.0 \text{ \AA}$ for each of the cations. The smaller the dipole, the more symmetric the polyhedron.

ration of simulations 2 and 3 are shown in Fig. 10. It shows that as the La ions cluster, the

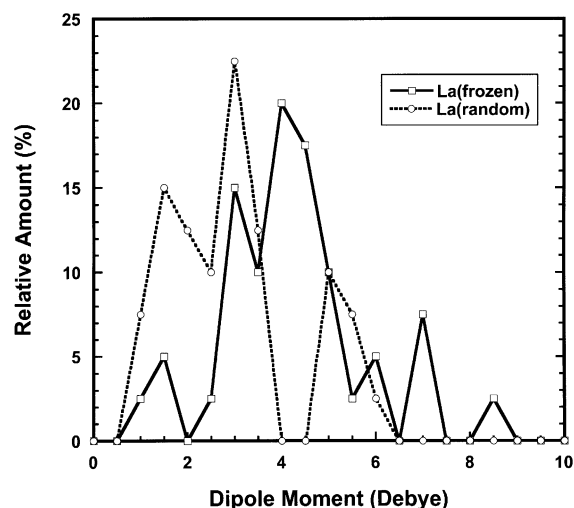


Fig. 10. The molecular dipole moment distribution of simulation 2 and 3. Note that for the case where the La ions are kept frozen the dipole moments are larger. The symmetry of the polyhedron increase when the La ions cluster.

dipole moment of the La polyhedra decrease. This implies that the La polyhedra become more symmetric. This shows that there is a strong effect on the network structure about the La cations.

The total configuration energy of the system as a function of La–La separation is given in Fig. 11. This is for a system containing only two La cations, with the same number of Na and La cations, with corresponding O anions, in the previous simulation. This mixture corresponds to a concentration of about 0.2 mol%. In these simulations, the La ions are held fixed while allowing the rest of the ions to go through the melt-quench glass-forming schedule. In this simulation six degrees of freedom were held fixed. Instead, it is preferable to use holonomic constraints such that only the distance between the ions is held fixed and otherwise the ions can translate and rotate in their constrained state. A more appropriate simulation approach to study this effect is reported elsewhere [15]. For this work, the idea was to determine which energy contributions were important for the separated versus the aggregated configurations. Each point on this path represents only one possible configuration of the glass for that separation

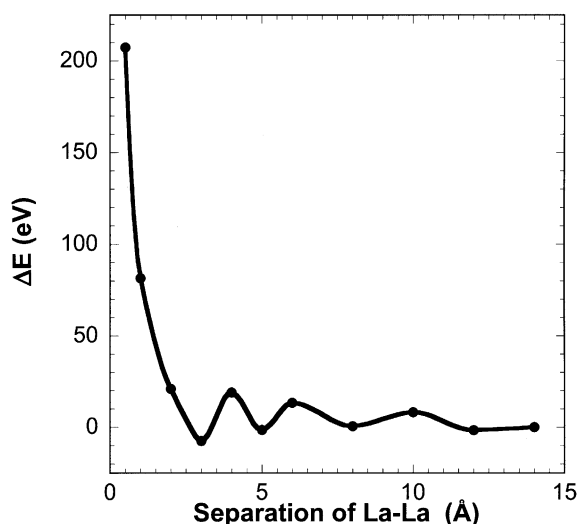


Fig. 11. The configurational energy of a glass containing only two La ions where the ion pair was held fixed at different distances, as indicated by the points on the graph.

distance of the La cations. The line is only a fit to the data points. The energy difference between the separated ions at 12 Å and the aggregated ions at about 3.5 Å is about 5 eV. Hence, this provides only a qualitative representation of the energies. However, it can be quantitatively partitioned as follows.

A profile of the O atom site energy for the Na₃S₄ composition is shown in Fig. 12. The distribution represents O atoms that have bonds to two Si atoms, usually referred to as bridging oxygen (BO) bonds; a bond to one Si atom with the other bond to a La atom, usually referred to as a non-bridging oxygen (NBO) bonds; and bonds associated with the formation of La–O–La free oxygen atoms. From this distribution, it is expected that changing an O atom from an NBO to a free oxygen costs about 1 eV in energy. Hence, in Fig. 11 only about 1 eV of the 5 eV difference between the separated and aggregated states is due to bond enthalpies. The rest of the energy appears to be due to structural response of the network. Note also that entropy contributions are not included in this calculation. This will be addressed in the next paper of this series [15].

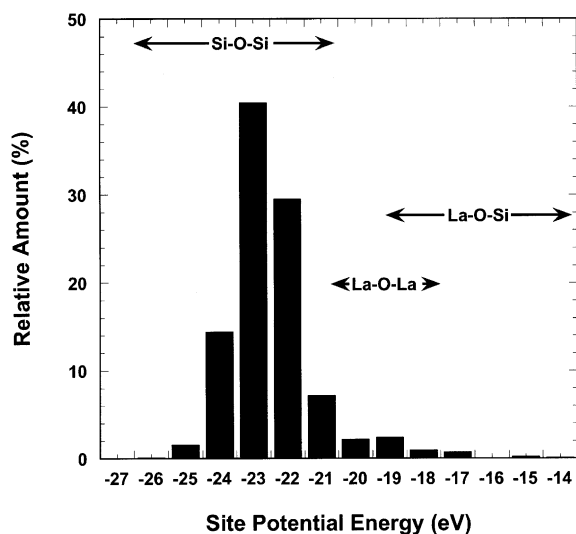


Fig. 12. The energy distribution of bond types formed by the La and Si cations with O atoms. The lowest in energy is that of the Si–O–Si type. The highest in energy is that of the Si–O–La type. The energy of forming La–O–La bonds is higher than that of Si–O–Si type and only about 1 eV more stable than that of the Si–O–La type.

4. Discussion

In the high-temperature melt of the simulations, there is sufficient diffusion of all the ions such that a statistical distribution of the ions is expected. Upon cooling, the distribution of the ions is expected to evolve such that the free energy of the system is minimized. The extreme case being that the system could crystallize. It is not clear when the La clustering sets in during the melt-quench glass-forming process since clusters are statistically predicted to appear and disappear when the La ions freely diffuse at the higher temperatures. As was shown in the above simulations, the initial configurations do not play a significant role since the high-temperature melt has the tendency to erase the memory of the initial configuration. It is clear that at some temperature the clusters are locked in place because they are energetically favorable. Just above this temperature, the melt is expected to have reached an equilibrium distribution and, hence, the structure observed at room temperature is only one of many possible equilibrium distributions for the glassy state. Thus, we

can ascertain that clustering is favored if an equilibrium statistical distribution of the ions is achieved at high temperature.

Having established that clustering of La ions sets in at a high temperature, the persistence of clustering was examined by annealing experiments in which the La ions were uniformly distributed. It was shown that when the La ions were frozen in a uniform configuration all of the other ions adjusted to this new environment. What is surprising is that the structural environments of the Si, Na and O ions appear to be similar whether the La ions are distributed uniformly or are clustered. Hence, at first glance it seems as though the network has no influence on the distribution of the La ions. There are, however, significant differences in the total configurational energy and coordination structure about the La cations as revealed by the degree of symmetry, as measured by the dipole moments of the cation coordination polyhedron. These differences are also reflected in the La–O PDF and to a lesser degree in the bond angle distributions.

The annealing studies reveal that the network has a strong influence on aggregation of the La ions. There is no reason to believe that the isolated La polyhedra have a strong attraction to one another given that there is a relatively small gain in energy in forming La–O–La linkages. Thus, the majority of the energy difference is due to the relaxation of the surrounding network. The network structure behaves as if to form solvation cavities in which the La cations can be accommodated. These cavity-like structures were previously found in simulations of Er^{3+} cation in silica and soda silicate glasses [16]. The individual ion cavities then have an energy associated with having a solvation shell form about the ion. In the case of pure silica and soda silicate networks coalescence of at least a few single La ion cavities can be energetically favorable if the system favors forming a larger cavity. In this case, if the cavities are in close proximity to one another, they will coalesce thus driving the La cations to aggregate. In the other extreme, there can be an upper limit due to an increase in energy when several single cavities coalesce to form a much larger one. If instead the energy to coalesce several cavities is favorable, the

latter can lead to macroscopic clustering associated with nanocluster formation, crystal nucleation and phase separation. This will be explored in future work.

The microheterogeneity of alkali silicate is well known and has been corroborated by experiment and MD simulations [9]. In the case of pure alkali silicate glass, where the alkali ions are located in percolation channels, the *modified random network* (MRN) model has been used to describe the structural distribution of network forming cations [17]. According to MRN model, the alkali ions are heterogeneously distributed in silicate glasses, forming alkali rich regions. In the above simulations the Na ions cluster amongst each other and may effect the clustering of La ions. However, as shown in paper I, the La silicate glass without Na still shows clustering of the La ions, thus, it is believed that the La ions have their own tendency to cluster. The presence of Na ions in the simulated glasses may enhance the clustering of La cations, but further investigation is required. It does, however, appear that the presence of Na ions does lead to a redistribution of the clustered La ions to a slightly more homogenous distribution. This is indicated by the disappearance of the La–La PDF peak at 4.8 Å and the shift of the first peak at 3.8–4.5 Å in going from the SiLa4 mixture to the NaLa4 mixture. The depolymerization of the silica network by addition of Na ions has the effect of allowing the network structure to adapt more readily to the presence of La ions. Thus, the network relaxation forces that drive the clustering of the La ions are changed.

In a real glass, it is not well understood how to define atomistic clustering or the microheterogeneity at a larger scale. It is unclear if the MRN model would be valid for the La ion heterogeneous structure formation, since the overall ion distribution is distinct from that of the alkali ions. However, MD simulations provide a means of identifying clustering of the La atoms into groups that contain La dimers, trimers, and so on by direct La–O–La linkages. The formation of such linkages is detectable by experiment, but how they are distributed throughout a glass is currently only accessible through computer simulation. In general, the MD simulations show that La³⁺ ions

appear to form quasi-molecular complexes that maintain their own environments consistent with the general phenomenon in glass systems where network modifiers form their own environment.

The energetic preference of clustering as opposed to uniform distribution of the La polyhedra could be, to some extent, a consequence of the potential model. One can imagine developing a set of potential parameters based on a uniform distribution of La cations that would favor such a distribution. However, most potentials used for amorphous systems are fit to crystal states and otherwise tested to reproduce crystal properties. Hence, the importance of developing transferable potentials, and their validation, based on the pure components of a system is further emphasized by these results. In addition, the role of charge transfer, polarizability of the O [18] and La ions, and the presence of water in these glasses must be studied further. These are topics of ongoing and future work that address the issues of sensitivity of simulations results on potential models [19,20].

5. Conclusion

The persistence of clustering of La ions in a soda silicate glass was studied by MD simulations. Clustering was found to occur when simulations were run at high temperatures where significant diffusion occurred such that the memory of the initial configuration was erased. In contrast, the initial configuration was found to have a strong influence on the degree of clustering for annealing temperatures where diffusion was insufficient to erase the memory of the initial configuration. Under such conditions, clustering may manifest simply because of a statistical distribution of the ions in the initial configuration. The role of network relaxation in stabilizing the formation of clusters was determined by annealing experiments where clustering occurred even though the temperatures were not high enough to produce a random distribution due to diffusion. Hence, clustering is found to be driven by network relaxation and by the energetics of forming La–O–La linkages.

Acknowledgements

This work was supported by the Division of Chemical Sciences, Office of Basic Energy Sciences, Department of Energy. This research was performed in the William R. Wiley Environmental Molecular Sciences Laboratory, a national scientific user facility sponsored by the Department of Energy, Office of Biological and Environmental Research and located at Pacific Northwest National Laboratory. Pacific Northwest Laboratory is operated for the Department of Energy by Battelle.

References

- [1] T. Schaller, J.F. Stebbins, M.C. Wilding, *J. Non-Cryst. Solids* 243 (1999) 146.
- [2] B. Park, H. Li, L.R. Corrales, *J. Non-Cryst. Solids* 297 (2002) 220.
- [3] M.C. Wilding, A. Navrotsky, *J. Non-Cryst. Solids* 254 (2000) 238.
- [4] E.M. Larson, A.J.G. Ellison, F.W. Lytle, A. Navrotsky, R.B. Greegor, *J. Non-Cryst. Solids* 130 (1991) 260.
- [5] A.J.G. Ellison, P.C. Hess, *J. Non-Cryst. Solids* 127 (1991) 247.
- [6] V. McGahay, M. Tomozawa, *J. Non-Cryst. Solids* 159 (1993) 246.
- [7] C. Huang, A.N. Cormack, *J. Chem. Phys.* 95 (1991) 3634.
- [8] B. Park, A.N. Cormack, *J. Non-Cryst. Solids* 255 (1999) 112.
- [9] G.N. Greaves, K.L. Ngai, *Phys. Rev.* 52 (1995) 6358, and references therein.
- [10] W. Smith, T.R. Forester, *J. Mol. Graphics* 14 (1996) 136.
- [11] P.P. Ewald, *Ann. Phys. (Leipzig)* 64 (1921) 253.
- [12] B. Vessal, *J. Non-Cryst. Solids* 177 (1994) 103.
- [13] See for example M.P. Allen, D.J. Tildesley, *Computer Simulation of Liquids*, Clarendon, Oxford, 1989.
- [14] D.J. Evans, G.P. Morriss, *Phys. Lett.* 98A (1983) 433; D.J. Evans, G.P. Morriss, *Chem. Phys.* 77 (1983) 63.
- [15] L.R. Corrales, B. Park, *J. Non-Cryst. Solids* 311 (2002) in press.
- [16] G. Cormier, J.A. Capobianco, A. Monteil, *J. Non-Cryst. Solids* 152 (1993) 225.
- [17] G.N. Greaves, *J. Non-Cryst. Solids* 71 (1985) 203.
- [18] J.A. Duffy, *J. Non-Cryst. Solids* 297 (2002) 275–284.
- [19] S.M. Kathmann, G.K. Schenter, B.C. Garrett, *J. Chem. Phys.* 116 (2002) 5046.
- [20] X.L. Yuan, A.N. Cormack, *J. Non-Cryst. Solids* 283 (2001) 69.